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Windows PowerShell
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W9D4_LangChain_Hybrid_Agents> python main.py

#####
W9D4 - LANGCHAIN HYBRID AGENT SYSTEMS
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TASK 1: LANGCHAIN CHAIN
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Test 1
Input: What is artificial intelligence?
Output: Artificial intelligence (AI) refers to the development of computer systems that can perform tasks that typically require human intelligence, such as learning, problem-solving, and decision-making. AI systems use algorithms and data to simulate human-like intelligence, enabling them to automate tasks, analyze data, and make predictions. This technology has numerous applications in various industries, including healthcare, finance, and transportation.

Test 2
Input: What is machine learning?
Output: Machine learning is a type of artificial intelligence that enables computers to learn from data and improve their performance on a specific task without being explicitly programmed. It involves training algorithms on large datasets to enable them to make predictions, classify objects, or make decisions. This allows machines to learn from experience and adapt to new situations.

Test 3
Input: What is LangChain?
Output: LangChain is an open-source framework for building scalable, modular, and maintainable blockchain applications. It provides a set of tools and libraries to simplify the development process, making it easier to integrate blockchain functionality into existing applications. LangChain enables developers to focus on building business logic rather than blockchain-specific code.

Test 4
Input: What is an AI agent?
Output: An AI agent is a software program that can perceive its environment, take actions, and learn from its experiences to achieve a specific goal. It can interact with its surroundings, make decisions, and adapt to new situations. AI agents are often used in applications such as robotics, game playing, and autonomous vehicles.

Test 5
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Test 5
Input: Why is Python useful for AI?
Output: Python is useful for AI due to its simplicity, readability, and extensive libraries, such as NumPy, pandas, and scikit-learn, which provide efficient data structures and algorithms for machine learning and data analysis. Additionally, Python's vast number of machine learning frameworks, including TensorFlow and Keras, make it an ideal choice for building and training AI models. Its ease of use and flexibility also make it a popular choice among data scientists and researchers.

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TASK 2: CONVERSATION BUFFER MEMORY
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C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W9D4_LangChain_Hybrid_Agents\main.py:79: LangChainDeprecationWarning: The class 'ConversationBufferMemory' was deprecated in LangChain 0.3.1 and will be removed in 2.0.0. Use 'langchain.agents.create_agent' instead. For agents that need to remember prior interactions, use 'create_agent' with checkpointing or the 'Store' API. See https://docs.langchain.com/oss/python/langchain/short-term-memory and https://docs.langchain.com/oss/python/langchain/long-term-memory
  memory = ConversationBufferMemory(

Turn 1
User: My name is Sanjana.
Assistant: Nice to meet you, Sanjana! How can I assist you today?

Turn 2
User: I am an Information Science student.
Assistant: Nice to meet you, Sanjana! As an Information Science student, you're likely interested in exploring the intersection of technology and information. What specific area of Information Science are you interested in learning more about?

Turn 3
User: I am learning artificial intelligence.
Assistant: That's a fascinating field! As an Information Science student learning about artificial intelligence, you might be interested in exploring the applications of AI in various domains, such as natural language processing, computer vision, or machine learning. What specific aspects of AI are you currently focusing on, Sanjana?

Turn 4
User: I am currently working on a LangChain task.
Assistant: LangChain is a fascinating project. It's a framework for building custom data pipelines and workflows. What specific challenges or aspects of LangChain are you currently working on, Sanjana?

Turn 5
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Turn 5
User: What is my name and what am I currently learning?
Assistant: Nice to meet you again, Sanjana! You're currently learning about artificial intelligence, and you're working on a LangChain task.

--- FINAL STORED CONVERSATION HISTORY ---
Human: My name is Sanjana.
AI: Nice to meet you, Sanjana! How can I assist you today?
Human: I am an Information Science student.
AI: Nice to meet you, Sanjana! As an Information Science student, you're likely interested in exploring the intersection of technology and information. What specific area of Information Science are you interested in learning more about?
Human: I am learning artificial intelligence.
AI: That's a fascinating field! As an Information Science student learning about artificial intelligence, you might be interested in exploring the applications of AI in various domains, such as natural language processing, computer vision, or machine learning. What specific aspects of AI are you currently focusing on, Sanjana?
Human: I am currently working on a LangChain task.
AI: LangChain is a fascinating project. It's a framework for building custom data pipelines and workflows. What specific challenges or aspects of LangChain are you currently working on, Sanjana?
Human: What is my name and what am I currently learning?
AI: Nice to meet you again, Sanjana! You're currently learning about artificial intelligence, and you're working on a LangChain task.

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TASK 3: TWO-TOOL LANGCHAIN AGENT
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Agent Task 1
User: Search for information about LangChain.
Agent: LangChain is an open-source framework that enables developers to build applications using language models, tools, agents, and memory. It provides a set of tools and libraries that allow developers to integrate language models into their applications, making it easier to build conversational AI, text generation, and other natural language processing (NLP) applications.

LangChain's key features include:

1. Modular architecture: LangChain is designed to be modular, allowing developers to easily integrate different components and tools into their applications.
2. Language model integration: LangChain provides a simple way to integrate language models into applications, making it easy to
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ols into their applications.
2. Language model integration: LangChain provides a simple way to integrate language models into applications, making it easy to use pre-trained models or train custom models.
3. Agent-based architecture: LangChain uses an agent-based architecture, which allows developers to define custom agents that can interact with language models and other components.
4. Memory management: LangChain provides a memory management system that allows developers to manage the state of their applications, making it easier to build complex applications.

LangChain has a wide range of applications, including:

1. Conversational AI: LangChain can be used to build conversational AI applications, such as chatbots and virtual assistants.
2. Text generation: LangChain can be used to generate text, such as articles, stories, and product descriptions.
3. Content moderation: LangChain can be used to moderate online content, such as detecting spam or hate speech.
4. Language translation: LangChain can be used to translate text from one language to another.

Overall, LangChain is a powerful framework that enables developers to build a wide range of applications using language models and other NLP tools.

Agent Task 2
User: Calculate 125 * 8 + 50.
Agent: The result of the calculation 125 * 8 + 50 is 1050.

Agent Task 3
User: Search for information about Ollama and explain it briefly.
Agent: Ollama is an open-source, local execution framework for language models. It enables users to run large language models on their own computers, without relying on cloud-based services or requiring significant computational resources. This allows for more control over data privacy, security, and performance. Ollama is particularly useful for researchers, developers, and individuals who want to work with large language models without incurring costs or relying on internet connectivity.

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W9D4 COMPLETED
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(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W9D4_LangChain_Hybrid_Agents> pip freeze > requirements.txt
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W9D4_LangChain_Hybrid_Agents> type requirements.txt
annotated-types==0.8.0
anyio==4.15.1
certifi==2026.7.22
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anyio==4.15.1
certifi==2026.7.22
charset-normalizer==3.5.1
distro==1.9.0
greenlet==3.5.5
h11==0.16.0
httpcore==1.0.9
httpcore2==2.12.0
httpx==0.28.1
httpx2==2.12.0
idna==3.19
jsonpatch==1.33
jsonpointer==3.1.1
langchain==1.4.0
langchain-classic==1.0.8
langchain-core==1.6.2
langchain-ollama==1.1.0
langchain-protocol==0.0.19
langchain-text-splitters==1.1.2
langgraph==1.2.11
langgraph-checkpoint==4.2.0
langgraph-prebuilt==1.1.0
langgraph-sdk==0.4.4
langsmith==0.12.4
ollama==0.6.2
orjson==3.12.0
ormsgpack==1.12.2
packaging==26.3
pydantic==2.13.5
pydantic_core==2.46.5
PyYAML==6.0.3
requests==2.34.2
requests-toolbelt==1.0.0
sniffio==1.3.1
SQLAlchemy==2.0.52
tenacity==9.1.4
truststore==0.10.4
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httpx2==2.12.0
idna==3.19
jsonpatch==1.33
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requests-toolbelt==1.0.0
sniffio==1.3.1
SQLAlchemy==2.0.52
tenacity==9.1.4
truststore==0.10.4
typing-inspection==0.4.4
typing_extensions==4.16.0
urllib3==2.7.0
uuid_utils==0.17.1
websockets==16.1.1
xxhash==4.0.1
zstandard==0.25.0
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W9D4_LangChain_Hybrid_Agents> notepad README.md
(venv) PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W9D4_LangChain_Hybrid_Agents>
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